



Combinatorics Problem Solving

- Raghav Goel



Problem 1

You are given an integer array `nums` and a positive integer `k`. Return the sum of the **maximum** and **minimum** elements of all **subsequences** of `nums` with **at most** `k` elements.

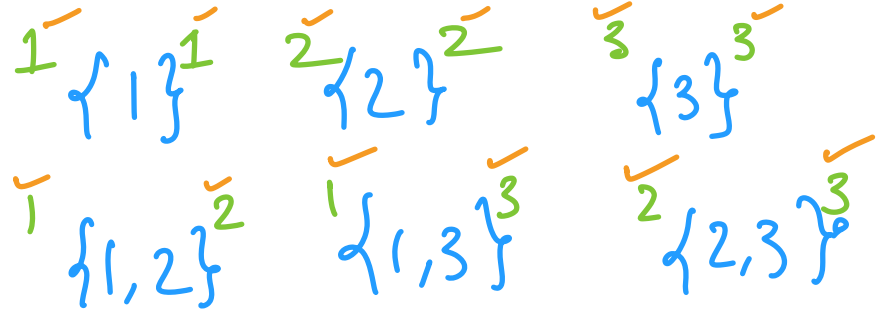
Since the answer may be very large, return it **modulo** $10^9 + 7$.

Link:

<https://leetcode.com/problems/maximum-and-minimum-sums-of-at-most-size-k-subsequences>

nums = [1, 2, 3]

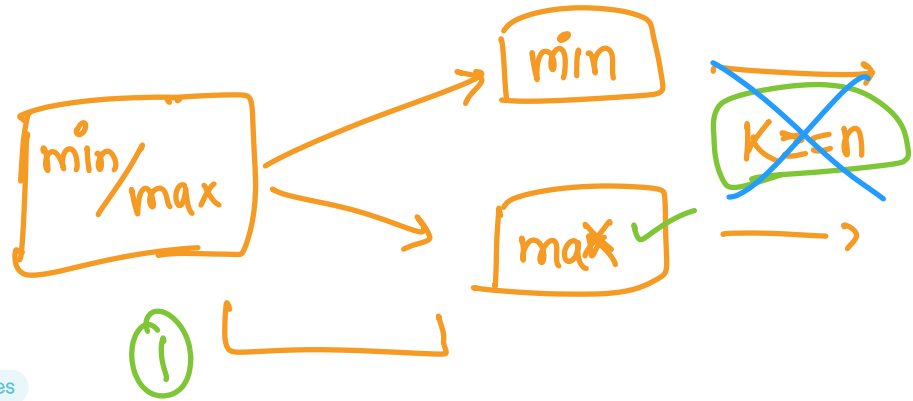
K=2



Sum of max.

Sum of min.

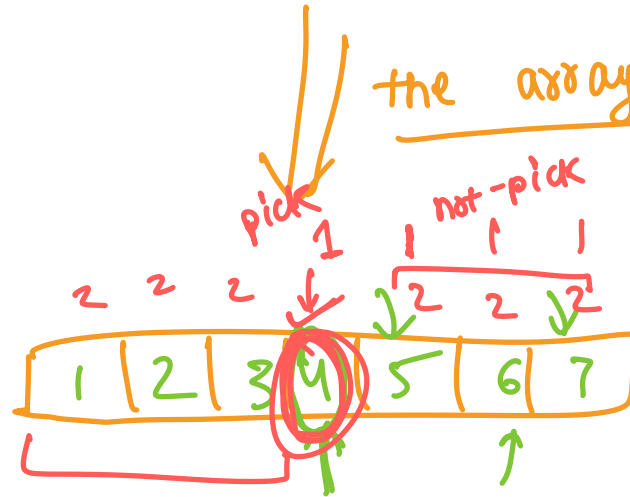
→ 24





sum of maximums of
all subsequences

pick
not-pick



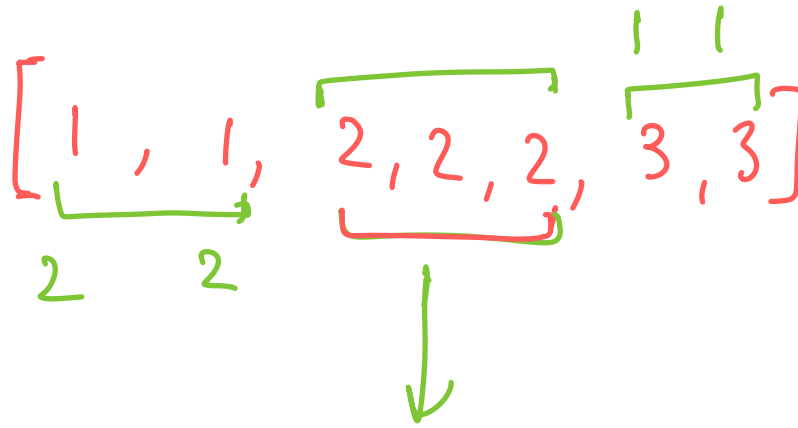
the array wouldn't have duplicates

unique elements

$$2^3$$

$$2^{n-1}$$

cnt_small



find the no. of
subsequences
where 2 is the
max. element

$$2^{\text{cnt_small}} * \left({}^3C_1 + {}^3C_2 + {}^3C_3 \right) \rightarrow 2^3 - 1$$

$$\underline{\underline{1 \leq K \leq \min(70, n)}}$$



$$\underline{\underline{1 \text{ to } K}}$$

size of subsequence should be 5

$a[i]$

cnt_small

✓ some

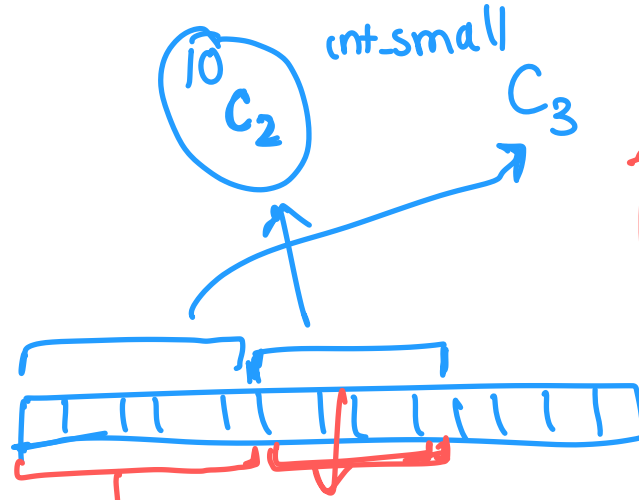
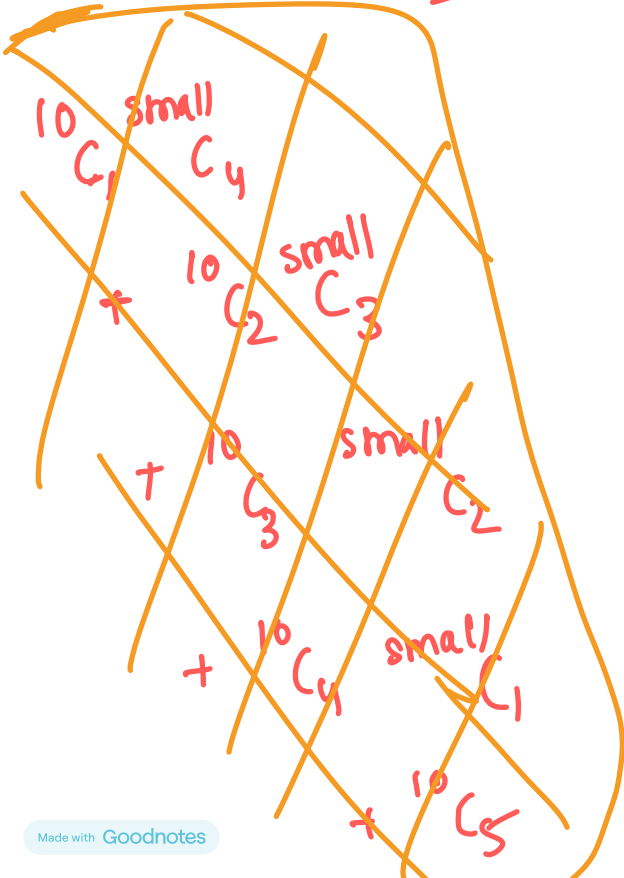
occurrences of $a[i]$

and these of the elements
should be smaller

$$sz = \underline{\underline{5}}$$

$$freq \rightarrow \underline{\underline{2}}$$

$$freq[a[i]] = 10$$



$$(2^{10} - 1)(2^{cnt_small})$$

$$\sum_{j=1}^j (10 C_j * cnt_small C_{sz-j})$$

$[sz = 1, 2, \dots, K] \rightarrow 20 \rightarrow O(K)$

for ($i=0; i < n; i++$) $\rightarrow$ (unique)

for $j=1, \dots, \text{freq}[a[i]]$

$ans += a[i] * \text{freq}[a[i]] * C_j$

cnt-small

C_{sz-1}

$O(1)$

fact[0] --- fact[N]
invfact[0] --- invfact[N]

runtime
 $(N+2) C_5$



Problem 2

You are given a 2D integer array, `queries`. For each `queries[i]`, where `queries[i] = [ni, ki]`, find the number of different ways you can place positive integers into an array of size n_i such that the product of the integers is k_i. As the number of ways may be too large, the answer to the *i*th query is the number of ways modulo $10^9 + 7$.

Return an integer array `answer` where `answer.length == queries.length`, and `answer[i]` is the answer to the *i*th query.

Link:

<https://leetcode.com/problems/count-ways-to-make-array-with-product>

Q queries

Q=1

$[n, k]$

no. of arrays of size n
where product of
all elements is k .

$n=5$

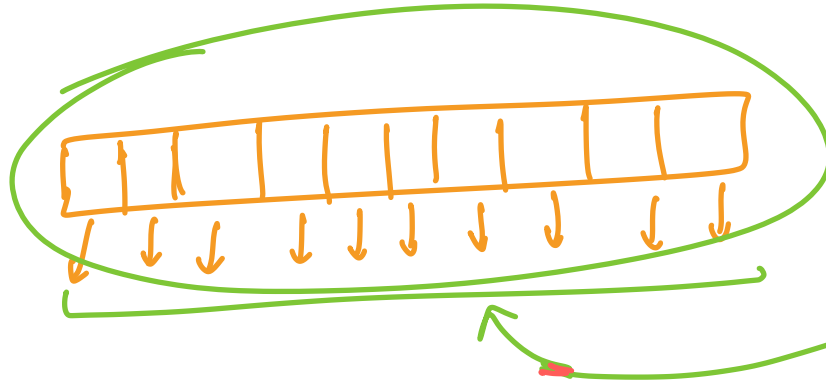
$k=6$

$[6, 1, 1, 1, 1]$

$[1, 6, 1, 1, 1]$

$[2, 3, 1, 1, 1]$

$$K = \begin{matrix} s_1 & s_2 & s_3 & s_4 \\ p_1 & p_2 & p_3 & p_4 \end{matrix}$$



$$K = 12 = 2 \times 2 \times 3$$

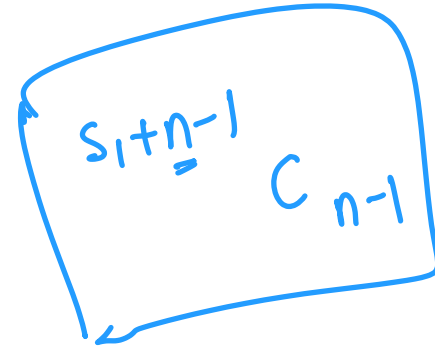
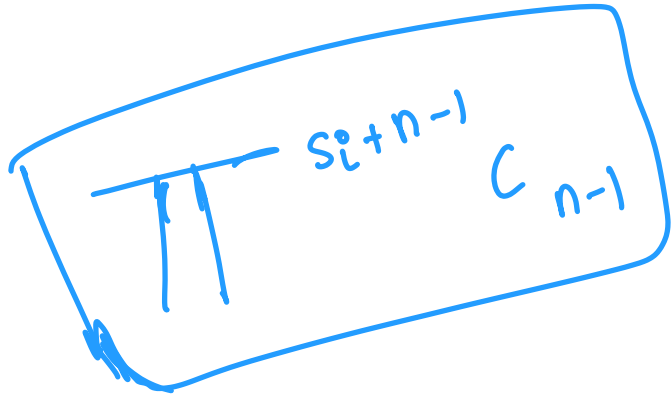
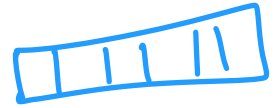
Diagram illustrating the factorization of $K = 12$ into $2 \times 2 \times 3$. Three green brackets group the factors. Red arrows point from each bracket to the letter n below. A red box contains n^3 .

$$[1, 1, 1, 1, 1, 1, 1, 1] \quad 2, 2, 3$$

$$\underline{\underline{[1, 1, 1, 1]}}$$

$K = \underbrace{P_1^{s_1}}_{\text{independent}} \underbrace{P_2^{s_2}} \underbrace{P_3^{s_3}} \dots$

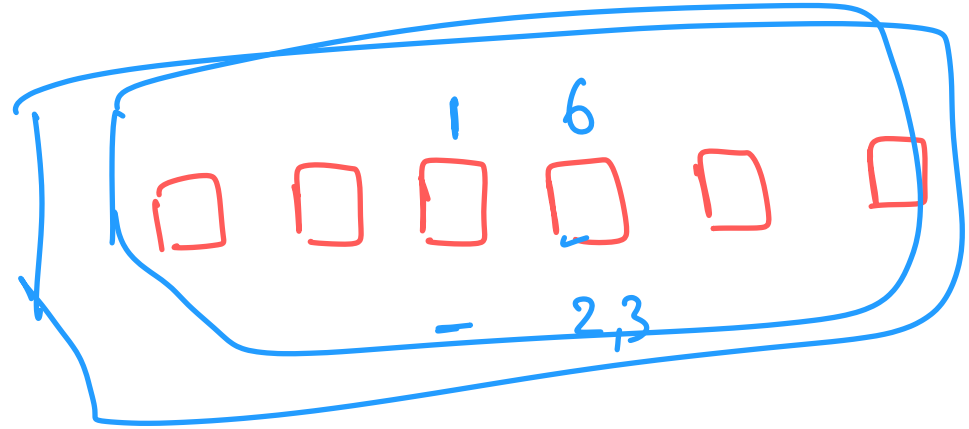
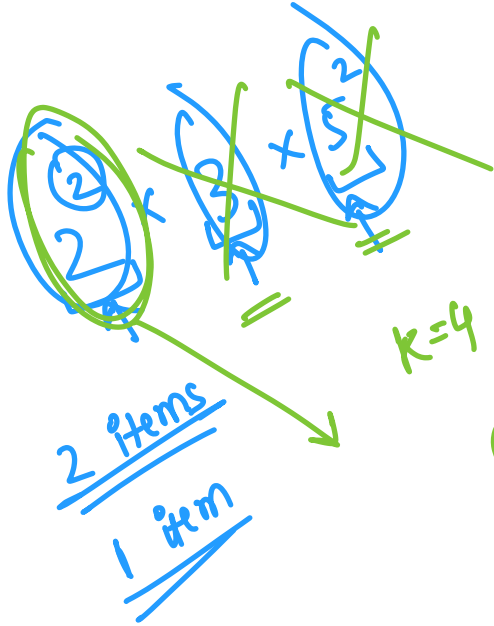
s_1 identical items
 n boxes



$n, k \Rightarrow$

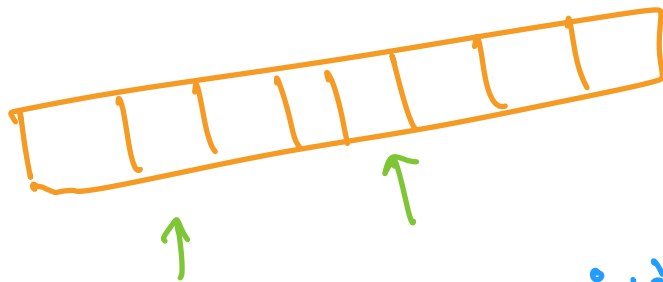
$k = 2 \times 2 \times 3$

$[1, 1, 1, \dots, 1]$



$f(n)$

$2 + n - 1 = c_{n-1}$



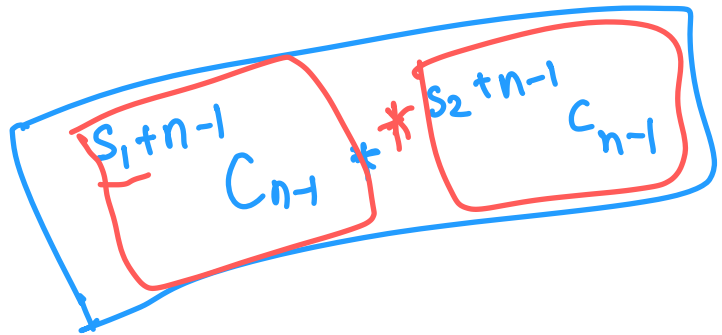
independent

distribution of 2
distribution of 3

are independent

$P_1 \rightarrow s_{1+n-1} \quad C_{n-1}$

$P_2 \rightarrow s_{2+n-1} \quad C_{n-1}$



Stars & Bars

Hard

$$k=30$$

$$n=2$$

$$k = \textcircled{2} \times 3 \times 5$$

8

$$\begin{aligned} [1,1] &\rightarrow [2,1] \\ &\searrow [1,2] \end{aligned}$$



Problem 3

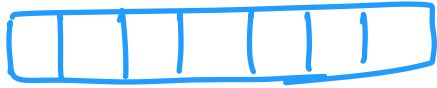
Hard



You are given an integer array `nums` and a **positive** integer `k`.
Return the sum of the **maximum** and **minimum** elements of all **subarrays** with **at most** `k` elements.

Link:

<https://leetcode.com/problems/maximum-and-minimum-sums-of-at-most-size-k-subarrays>



- ① $\Sigma \text{ max}$ ✓
- ② unique ✓
- ③ $K=n$ ✓

$K=n$ ✓

sum of max/min over all
subarrays of size atmost K

max ✓

unique elements ✓

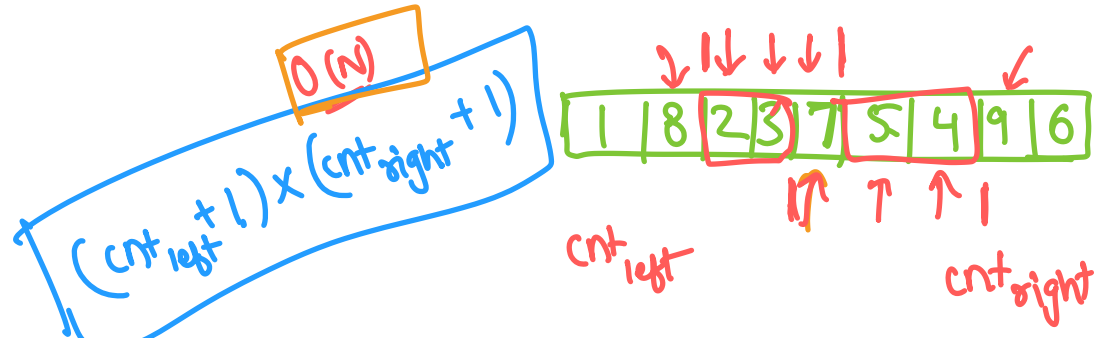
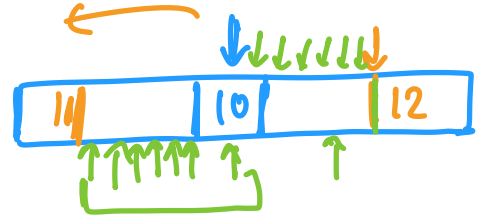


✓ $\Sigma \max$ over all subarrays
all elements are unique

no. of subarrays

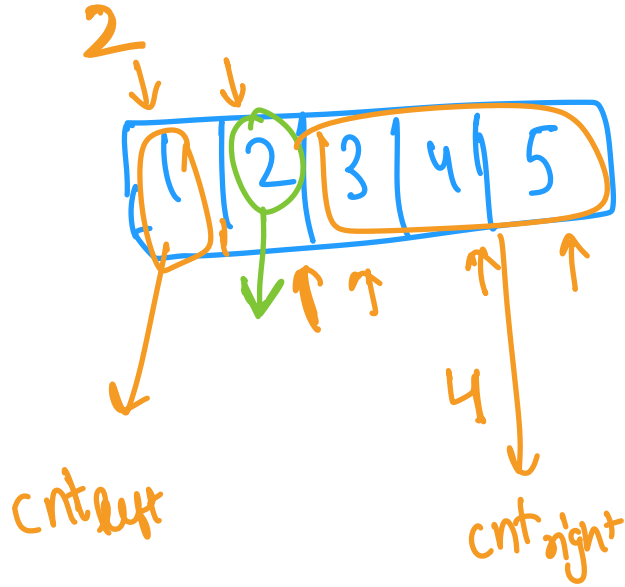
where $a[i]$ is the maximum element

$O(n)$ monotonic stack
pre compute



$O(n)$
 $(cnt_{left} + 1) \times (cnt_{right} + 1)$

$2 * 4$

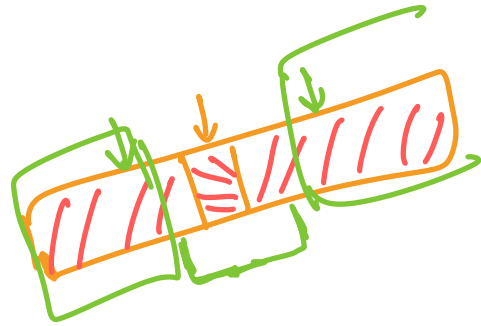


$\{1, 2\} \quad \{2\}$

$\{2, 3\} \quad \{2, 3, 4\} \quad \{2, 5\}$

$\{1, 2, 3\} \quad \{1, 2, 3, 4\}$
 $\{1, 2, 3, 4, 5\}$

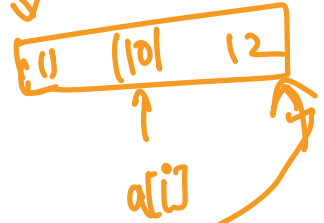
$\sum$ max. of all subarray]



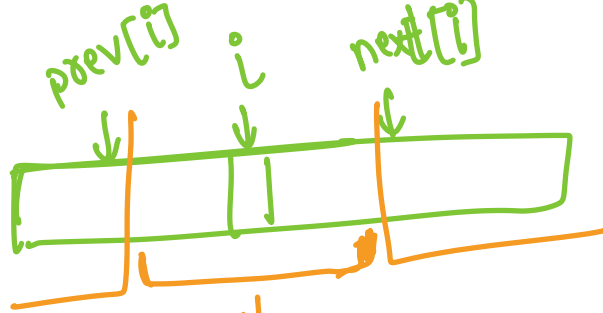
$a[i]$

cnt_{left}, cnt_{right}

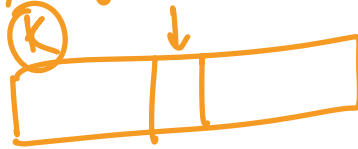
prev[i]
next[i]



$$\sum_{i=0}^{n-1} \underline{a[i]} \times (\text{cnt}_{\text{left}} + 1) \times (\text{cnt}_{\text{right}} + 1)$$



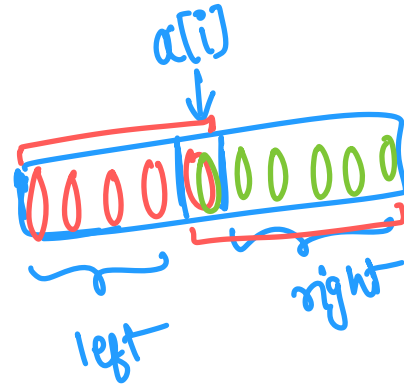
countSubarray(cnt_{left}, cnt_{right})



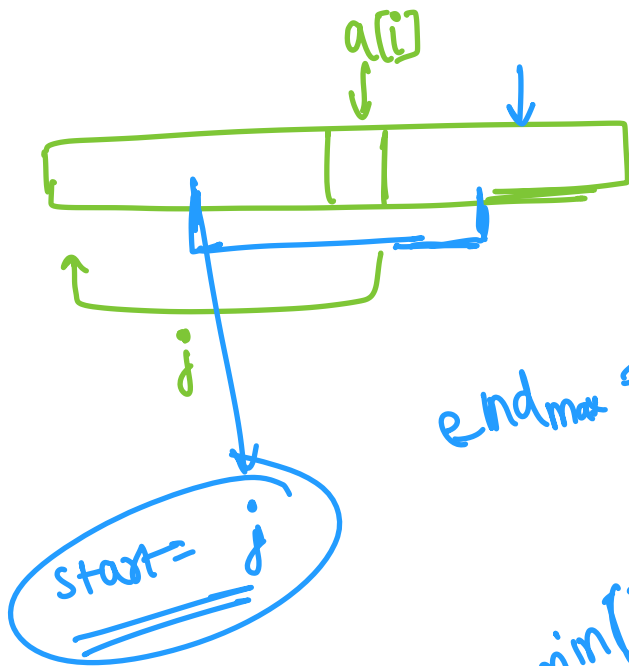
$[1, 2, 3, \boxed{4}, 5]$
 0 1 2 2

 size ≤ 3

countSubarrays (left, right, K)



TUE



$$\text{endmax} = j + k - 1$$

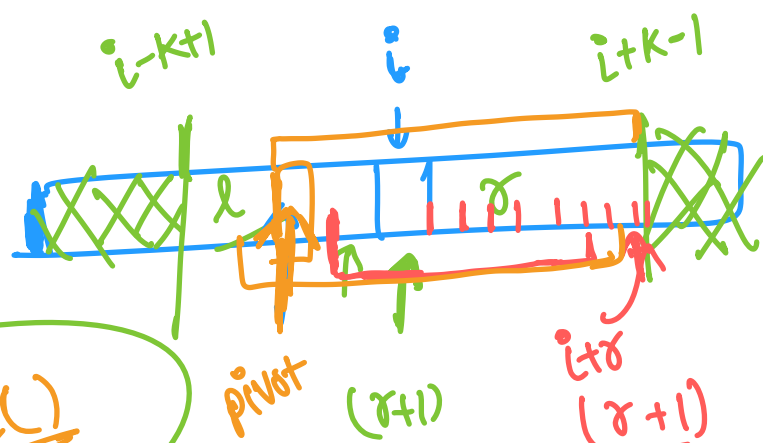
~~$[i, i+1, i+2, \dots, \min(j+k-1, \text{next}[i]-1)]$~~

~~$\text{endmax} \leq \text{next}[i]$~~

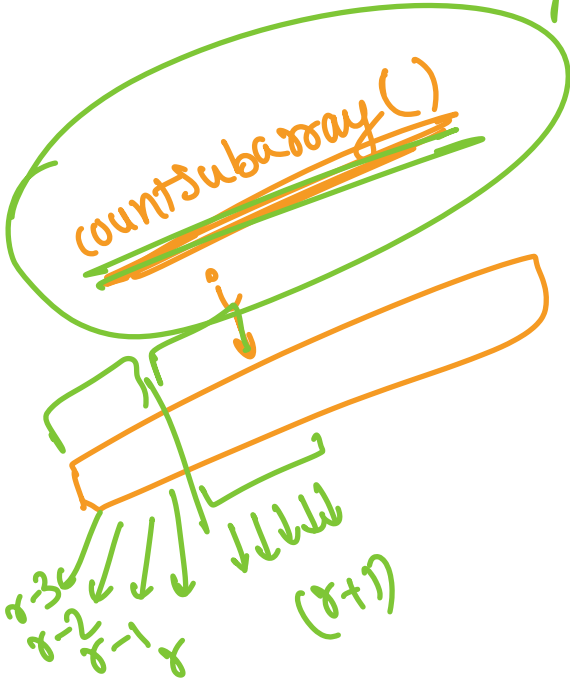
~~$\text{end} \geq i$~~

~~$\text{end} \leq \text{next}[i]$~~

~~$\text{end} \leq j+k-1$~~



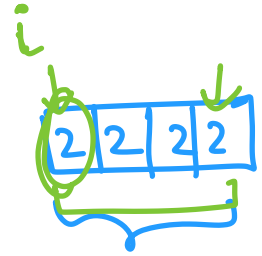
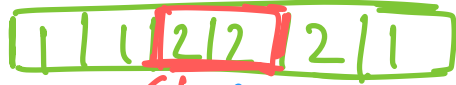
no. of subarrays in which
the index i is
included



start = j

$$\begin{aligned} \text{end} &\leq j+k-1 \\ \text{end} &\leq i+r \end{aligned}$$

$\text{end} \geq i$



left $\rightarrow \geq$
right $\rightarrow \leq$

should be counted only once

for each $a[i]$, find the number
of subarrays where $a[i]$ is
the first occurrence of the
max. element

max.

min.

[... , 0, ...]
↑
left

15
↑
[1, 2, 3, 4, 5]

↓
[-1, -2, -3, -4, -5]
-15

solve_for_max(nums, k)

for each x in nums:
 $x = -x$

Σ minimums

[- solve_for_max(nums, k)]

Problem 4

Homework



You are given a string s and an integer k .

A **k-subsequence** is a **subsequence** of s , having length k , and all its characters are **unique**, i.e., every character occurs once.

Let $f(c)$ denote the number of times the character c occurs in s .

The **beauty** of a **k-subsequence** is the **sum** of $f(c)$ for every character c in the k-subsequence.

Return an integer denoting the number of k-subsequences whose **beauty** is the **maximum** among all **k-subsequences**. Since the answer may be too large, return it modulo $10^9 + 7$.

Link:

<https://leetcode.com/problems/count-k-subsequences-of-a-string-with-maximum-beauty>